

Logic Exam Responses

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Introduction

These are my solutions to the exam problems. If my style should seem a little odd, it is probably because in my day job I am a software engineer / manager. I aim for rigor, but I also favor clarity, brevity, and structured thinking – too much time writing code and fixing other people’s code. Apologies in advance if this isn’t the response structure from the formal class.

Part I: Valid or Invalid

1. $R \rightarrow \neg P, Q, Q \rightarrow (P \vee \neg S) \vdash S \rightarrow \neg R$

Valid.

Proof sketch: From Q and $Q \rightarrow (P \vee \neg S)$ infer $P \vee \neg S$. Assume S . Then P must hold (since $\neg S$ contradicts S). If we further assume R , then by $R \rightarrow \neg P$ we derive $\neg P$, contradiction. Thus $\neg R$. Hence $S \rightarrow \neg R$.

2. $\neg P \leftrightarrow Q, P \rightarrow R, \neg R \vdash \neg Q \leftrightarrow R$

Valid.

Proof sketch: From $P \rightarrow R$ and $\neg R$ we obtain $\neg P$. Then using $\neg P \leftrightarrow Q$ we get Q . Hence $\neg Q$ is false and R is false, making $\neg Q \leftrightarrow R$ true. A natural deduction proof shows both directions ($\neg Q \rightarrow R$) and ($R \rightarrow \neg Q$).

3. $\vdash P \leftrightarrow (P \vee (P \wedge Q))$

Valid.

Proof sketch: $P \vee (P \wedge Q)$ simplifies logically to P . So this is equivalent to $P \leftrightarrow P$, a tautology.

Part II: Truth Tables

1. $(P \vee Q) \leftrightarrow (P \wedge Q) \vdash P \leftrightarrow Q$

Valid.

Truth table check: The premise is true only when P and Q have the same truth value (both true or both false). In those rows, $P \leftrightarrow Q$ is also true. Hence the sequent is valid.

2. $P \rightarrow (Q \vee \neg R), R \wedge Q \vdash \neg P$

Invalid.

Counterexample: Let $P=T, Q=T, R=T$. Then $P \rightarrow (Q \vee \neg R)$ is true, $R \wedge Q$ is true, but $\neg P$ is false.

Part III: Proofs / Countermodels

1. $(\forall x Fx \rightarrow \forall x Gx) \vdash \exists x (Fx \rightarrow Gx)$

Valid.

Reasoning: If $\forall x Fx$ fails, then for some witness a , Fa is false, making $(Fa \rightarrow Ga)$ true. If $\forall x Fx$ holds, then by premise $\forall x Gx$ also holds, and any witness satisfies $Fx \rightarrow Gx$. Either way, an existential witness exists.

2. $\vdash \forall x \forall z Fxz \rightarrow \forall y \forall z Fyz$

Valid.

Reasoning: The renaming of bound variables shows that if F holds for all x and z , then it holds for all y and z .

3. $\exists x (Exa \vee Fx) \vee Fa, Eba \vdash Fa \vee Fb$

Invalid.

Countermodel: Domain $\{a, b\}$. Let Eba be true, Fa false, Fb false. Then the premises are true but $Fa \vee Fb$ is false.

4. $(\exists x Ax \rightarrow \forall y (By \rightarrow Cy)), (\exists x Dx \rightarrow \exists y By) \vdash \exists x (Ax \wedge Dx) \rightarrow \exists y Cy$

Valid.

Reasoning: If $\exists x (Ax \wedge Dx)$ holds, then some witness c has both Ac and Dc . From this we infer $\exists x Ax$ and $\exists x Dx$. The premises then yield $\forall y (By \rightarrow Cy)$ and $\exists y By$, giving some witness d with Bd and Cd . Hence $\exists y Cy$.

5. $\exists x (Fx \wedge Gx), \exists x (\neg Fx \wedge \neg Gx) \vdash \exists x (Gx \wedge \neg Fx)$

Invalid.

Countermodel: Domain $\{a, b\}$. Assign $F(a) = T, G(a) = T; F(b) = F, G(b) = F$. Then the premises are true, but there is no x with $Gx \wedge \neg Fx$.

Exam Comments:

Overall, the results I arrived at seem consistent with standard reasoning. Some of these are provable, others fall to counterexamples. However, I remain skeptical of my own response to problem III.3. This the one where I am least confident. My countermodel convinces me it is invalid, yet there is something which is just odd about it. At this point, I rely on my countermodel – final answer as they say ...

Also, I attached my truth-tables after this section for review as needed.

Formatted in LaTeX, because code, math, and philosophy all deserve clean typesetting.